

Tamil Nadu - Grade 10

2016 - Mathematics

Maximum Marks: 100

SECTION – I

1. If $A = \{p, q, r, s\}$, $B = \{r, s, t, u\}$, then $A \setminus B$ is:

- (a) $\{a, q\}$
- (b) $\{t, u\}$
- (c) $\{r, s\}$
- (d) $\{p, q, r, s\}$

Answer: (a)

Solution: $\{p, q\}$

2. The 8th term of the sequence 1, 1, 2, 3, 5, 8 is:

- (a) 25
- (b) 24
- (c) 23
- (d) 21

Answer: (d)

Solution: 21

3. If $1 + 2 + 3 + \dots + n = k$ then $1^3 + 2^3 + 3^3 + \dots + n^3$ is equal to:

- (a) k^2
- (b) k^3
- (c) $\frac{k(k+1)}{2}$
- (d) $(k+1)^3$

Answer: (a)

Solution: k^2

4. The system of equations $x - 4y = 8$, $3x - 12y = 24$.

- (a) has infinitely many solutions
- (b) has no solution
- (c) has a unique solution
- (d) may or may not have a solution

Answer: (a)

Solution: has infinitely many solutions

$$a_1 = 1 \quad b_1 = -4 \quad c_1 = 8$$

$$a_2 = 3 \quad b_2 = -12 \quad c_2 = 24$$

$$\frac{1}{3} \quad \frac{1}{3} \quad \frac{1}{3}$$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

5. If $ax^2 + bx + c = 0$ has equal roots, then 'c' is equal to:

(a) $\frac{b^2}{2a}$

(b) $\frac{b^2}{4a}$

(c) $-\frac{b^2}{2a}$

(d) $-\frac{b^2}{4a}$

Answer: (b)

Solution: $\frac{b^2}{4a}$

$$\Delta = 0 \quad (\text{since roots are equal})$$

$$b^2 - 4ac = 0$$

$$c = \frac{b^2}{4a}$$

6. If $A = \begin{pmatrix} 4 & -2 \\ 6 & -3 \end{pmatrix}$, then A^2 is:

(a) $\begin{pmatrix} 16 & 4 \\ 36 & 9 \end{pmatrix}$

(b) $\begin{pmatrix} 8 & -4 \\ 12 & -6 \end{pmatrix}$

(c) $\begin{pmatrix} -4 & 2 \\ -6 & 3 \end{pmatrix}$

(d) $\begin{pmatrix} 4 & -2 \\ 6 & -3 \end{pmatrix}$

Answer: (d)

Solution: $\begin{pmatrix} 4 & -2 \\ 6 & -3 \end{pmatrix}$

7. The equation of a straight line parallel to y-axis and passing through the point $(-2, 5)$

(a) $x - 2 = 0$

(b) $x + 2 = 0$

(c) $y + 5 = 0$

(d) $y - 5 = 0$

Answer: (b)

Solution: $x + 2 = 0$

8. The equation of a straight line having slope 3 and y – intercept -4 is:

(a) $3x - y - 4 = 0$

(b) $3x + 7 + 4 = 0$

(c) $3x - y + 4 = 0$

(d) $3x + y + 4 = 0$

Answer: (a)

Solution: $3x - y - 4 = 0$

9. In $\triangle ABC$, DE is $\parallel BC$, meeting AB and AC at D and E . If $AD = 3$ cm, $DB = 2$ cm and $AE = 2.7$ cm, then AC is equal to:

(a) 6.5 cm

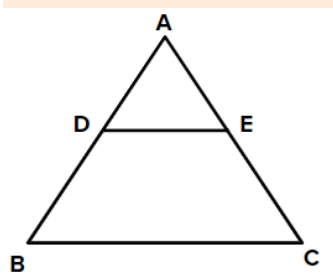
(b) 4.5 cm

(c) 3.5 cm

(d) 5.5 cm

Answer: (b)

Solution:



By B.P.T

$$\frac{AD}{DB} = \frac{AE}{EC}$$

$$\frac{3}{2} = \frac{2.7}{EC}$$

$$EC = 1.8$$

$$AC = 1.8 + 2.7$$

$$4.5 \text{ cm}$$

10. AB and CD are two chords of a circle which when produced to meet at a point P such that $AB = 5$ cm, $AP = 8$ cm and $CD = 2$ cm then $PD =$

- (a) 12 cm
- (b) 5 cm
- (c) 6 cm
- (d) 4 cm

Answer: (d)

Solution: 4 cm

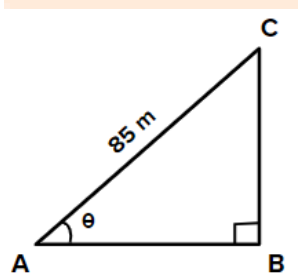
11. $(1 + \tan^2 \theta) \cdot \sin^2 \theta =$

- (a) $\sin^2 \theta$
- (b) $\cos^2 \theta$
- (c) $\tan^2 \theta$
- (d) $\cot^2 \theta$

Answer: (c)

Solution: $\tan^2 \theta$

12. In the adjoining figure, $\sin \theta = \frac{15}{17}$. Then BC =



- (a) 85 m
- (b) 65 m
- (c) 95 m
- (d) 75 m

Answer: (d)

Solution: 75 cm

13. The total surface area of a solid hemisphere of diameter 2 cm is equal to:

- (a) 12 cm^2
- (b) $12\pi \text{ cm}^2$
- (c) $4\pi \text{ cm}^2$
- (d) $3\pi \text{ cm}^2$

Answer: (d)

Solution: $3\pi \text{ cm}^2$

14. If t is the standard deviation of x, y, z , then the standard deviation of $x + 5, y + 5, z + 5$ is:

- (a) $\frac{t}{3}$
- (b) $t + 5$
- (c) t
- (d) xyz

Answer: (c)

Solution: t

15. A bag contains 5 black balls, 4 white balls and 3 red balls. If a ball is selected at random, the probability that it is not red is;

- (a) $\frac{5}{12}$
- (b) $\frac{4}{12}$
- (c) $\frac{3}{12}$
- (d) $\frac{3}{4}$

Answer: (d)

Solution: $\frac{3}{4}$

$$P(S) = 5 + 4 + 3 = 12$$

$$P(R) = \frac{3}{12} = \frac{1}{4}$$

$$P(R) = \frac{3}{12} = \frac{1}{4}$$

$$P(R') = 1 - \frac{1}{4} = \frac{3}{4}$$

16. Given $A = \{a, x, y, r, s\}$, $B = \{1, 3, 5, 7, -10\}$ verify the commutative property of set union.

Solution: $A \cup B = \{a, x, y, r, s, 1, 3, 5, 7, -10\}$

$$B \cup A = \{1, 3, 5, 7, -10, a, x, y, r, s\}$$

17. Let $A = \{5, 6, 7, 8\}$; $B = \{-11, 4, 7, -10, -7, -9, -13\}$ and $f = \{(x, y); y = 3 - 2x, x \in A, y \in B\}$

(i) Write down the elements of f .

(ii) What is the range?

Solution: $f = \{(5, -7), (6, -9), (7, -11), (8, -13)\}$

$$\text{range} = \{-7, -9, -11, -13\}$$

18. Find the 12th term of the A.P. $\sqrt{2}, 3\sqrt{2}, 5\sqrt{2}, \dots$

Solution:

$$a_0 = \sqrt{2}$$

$$d = 3\sqrt{2} - \sqrt{2} = 2\sqrt{2}$$

$$a_{12} = a_0 + 11d$$

$$= \sqrt{2} + 11 \times 2\sqrt{2}$$

$$= \sqrt{2} + 22\sqrt{2}$$

$$= 23\sqrt{2}$$

19. Simplify: $\frac{x^3}{x-2} + \frac{8}{2-x}$

Solution:
$$\begin{aligned} &= \frac{x^3}{x-2} - \frac{8}{x-2} \\ &= \frac{x^3(x-2) - 8(x-2)}{(x-2)^2} \\ &= \frac{x^3 - 2^3}{x-2} \\ &= \frac{(\cancel{x-2})(x^2 + 2x + 4)}{\cancel{x-2}} \\ &= x^2 + 2x + 4 \end{aligned}$$

20. If α and β are the roots of the equation $3x^2 - 6x + 4 = 0$, find the value of $\alpha^2 + \beta^2$.

Solution: $a = 3$ $b = -6$ $c = 4$

$$\alpha + \beta = \frac{-b}{a} = \frac{-(-6)}{3} = 2$$

$$\alpha\beta = \frac{c}{a} = \frac{4}{3}$$

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$= 2^2 - 2 \times \frac{4}{3}$$

$$= 4 - \frac{8}{3}$$

$$= \frac{4}{3}$$

21. Find the product of the matrices, if exists. $\begin{pmatrix} 2 & 9 & -3 \\ 4 & -1 & 0 \end{pmatrix} \begin{pmatrix} 4 & 2 \\ -6 & 7 \\ -2 & 1 \end{pmatrix}$

Solution: $\begin{pmatrix} 2 & 9 & -3 \\ 4 & -1 & 0 \end{pmatrix} \begin{pmatrix} 4 & 2 \\ -6 & 7 \\ -2 & 1 \end{pmatrix}$

$$= \begin{pmatrix} 8-54+6 & 4+63-3 \\ 16+6+0 & 8-7+0 \end{pmatrix}$$

$$\begin{pmatrix} -40 & 64 \\ 22 & 1 \end{pmatrix}$$

22. Find the values of x, y and z if $\begin{bmatrix} x+y \\ y+z \\ z-5 \end{bmatrix} = \begin{bmatrix} 7 \\ 9 \\ 0 \end{bmatrix}$

Solution: $z - 5 = 0$... (1)

$$z = 5$$

$y + z = 9$... (2)

$$y = 9 - 5$$

$$= 4$$

$x + y = 7$... (3)

$$x = 7 - 4$$

$$= 3$$

$$x = 3, y = 4, z = 5$$

23. If the centroid of a triangle is at $(1, 3)$ and two of its vertices are $(-7, 6)$ and $(8, 5)$ then find the third vertex of the triangle.

Solution: centroid = $\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}$

$$(1, 3) = \left(\frac{x_1 + -7 + 8}{3}, \frac{y_1 + 6 + 5}{3} \right)$$

$$1 = \frac{x_1 + 1}{3} \quad 3 = \frac{y_1 + 11}{3}$$

$$3 - 1 = x_1 \quad y_1 = 9 - 11$$

$$x_1 = 2 \quad y_1 = -2 \quad \rightarrow (2, -2)$$

24. Find the value of 'a', if the straight lines $5x - 2y - 9 = 0$ and $ay + 2x - 11 = 0$ are perpendicular to each other.

Solution:

$$2y = 5x - 9$$

$$y = \frac{5}{2}x - \frac{9}{2}$$

$$m_1 = \frac{5}{2}$$

$$ay = -2x + 11$$

$$y = \frac{-2}{a}x + \frac{11}{a}$$

$$m_2 = \frac{-2}{a}$$

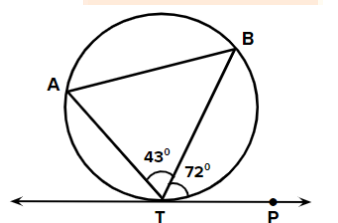
$\therefore$ lines are $\perp$

$$m_1 m_2 = -1$$

$$\frac{5}{2} \times \frac{-2}{a} = -1$$

$$\boxed{a = 5}$$

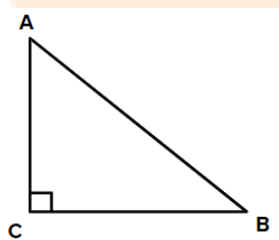
25. In the figure TP is a tangent to the circle. A and B are two points on the circle. If $\angle BTP = 72^\circ$ and $\angle ATB = 43^\circ$ find $\angle ABT$.



Solution: $\angle TAB = 72^\circ$ (because $\angle BTP = \angle BAT$)

$$\angle ABT = 65^\circ \text{ (because } \angle ABT = 180 - 72 - 43 \text{)}$$

26. If $\triangle ABC$ is right angled at 'C', then find the values of $\cos (A + B)$ and $\sin (A + B)$.



From $\triangle ABC$

$$\angle A + \angle B + \angle C = 180$$

$$\angle A + \angle B = 180 - 90$$

$$\angle A + \angle B = 90$$

$$\cos(A + B) = \cos 90 = 0$$

$$\sin(A + B) = \sin 90 = 1$$

27. Curved surface area and circumference at the base of a solid right circular cylinder are 4400 sq. cm and 110 cm respectively. Find its height and diameter.

Solution: C.S.A of cylinder = 4400 cm^2

$$2\pi rh = 4400 \quad \dots(i)$$

Circumference of cylinder = 110 cm

$$2\pi r = 110 \quad \dots(ii)$$

divide eq.(i) with eq.(ii)

$$\frac{2\cancel{\pi}rh}{2\cancel{\pi}r} = \frac{\overset{40}{\cancel{4400}}}{\cancel{110}}$$

$$h = 40 \text{ and } d = 2r = 35 \text{ cm}$$

28. If the coefficient of variation of a collection of data is 57 and its S.D. is 6.84, then find the mean.

Solution: $C_v = \frac{\theta}{x} \times 100$

$$C_v = 57 \quad \theta = 684$$

$$\bar{x} = \frac{\theta}{C_v} \times 100$$

$$= \frac{6.84}{57} \times 100$$

$$\bar{x} = 12$$

29. 20 cards are numbered from 1 to 20. One card is drawn at random. What is the Probability that the number on the card is?

(i) A multiple of 4

(ii) Not a multiple of 6

Solution:

$$P(S) = 20$$

$$P(4) = \{4, 8, 12, 16, 20\}$$

$$P(4) = \frac{5}{20} = \frac{1}{4}$$

$$P(6) = \{6, 12, 18\}$$

$$P(6) = \frac{3}{20}$$

$$P(6') = \frac{1-3}{20} = \frac{17}{20}$$

30.

(a) The central angle and radius of a sector of a circular disc are 180° and 21 cm respectively. If the edges of the sector are joined together to make a hollow cone, then find the radius of the cone.

OR

(b) Prove that: $\frac{1 + \sec \theta}{\sec \theta} = \frac{\sin^2 \theta}{1 - \cos \theta}$

Solution: circumference of hollow cone = length of sector

$$(a) 2\pi R = \frac{\theta}{360} \times 2\pi r$$

$$r = 21$$

$$R = \frac{180}{360} \times 21$$

$$= 10.5 \text{ cm}$$

$$(b) \text{ LHS} = \frac{1 + \sec \theta}{\sec \theta}$$

$$= \frac{1 + \frac{1}{\cos \theta}}{\frac{1}{\cos \theta}}$$

$$= 1 + \cos \theta$$

$$\text{LHS} = \text{RHS}$$

$$\text{RHS} = \frac{\sin^2 \theta}{1 - \cos \theta}$$

$$= \frac{1 - \cos^2 \theta}{1 - \cos \theta}$$

$$= \frac{(1 + \cos \theta)(\cancel{1 - \cos \theta})}{\cancel{1 - \cos \theta}}$$

$$= 1 + \cos \theta$$

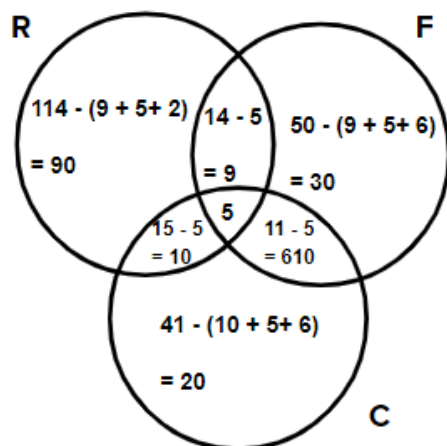
31. A radio station surveyed 190 students to determine the types of music they liked. The survey revealed that 114 liked rock music, 50 liked folk music and 41 liked classical music, 14 liked rock music and folk music, 15 liked rock music and classical music, 11 liked classical music and folk music, 5 liked all the three of music.

Find: (i) How many did not like any of the 3 types?

(ii) How many liked any two types only?

(iii) How many liked folk music but not rock music?

Solution:



- (i) No. of students who didn't like any of the three types = $190 - 170 = 20$
- (ii) No. of students who liked any of two types = $9 + 6 + 10 = 25$
- (iii) No. of students who liked folk but not rock = $30 + 6 = 36$

32. A Function $f: [-3, 7) \rightarrow \mathbb{R}$ is defined as follows: $f(x) = \begin{cases} 4x^2 - 1 & ; -3 \leq x < 2 \\ 3x - 2 & ; 2 \leq x \leq 4 \\ 2x - 3 & ; 4 < x < 7 \end{cases}$

Find: (i) $f(-2) - f(4)$

(ii) $\frac{f(3) + f(-1)}{2f(6) - f(1)}$

Solution:

$$f(-2) = 15; f(4) = 10$$

$$\left. \begin{aligned} f(3) &= 7; f(-1) = 3 \\ f(6) &= 9; f(1) = 3 \end{aligned} \right\}$$

$$\frac{f(3) + f(-1)}{2f(6) - f(1)} = \frac{10}{15} \text{ or } \frac{2}{3}$$

33. Find the sum to n terms of the series $7 + 77 + 777 + \dots$

Solution:

$$S_n = (1 + 11 + 111 + \dots \text{ n terms})$$

$$\begin{aligned}
 &= \frac{7}{9}(9 + 99 + 999 + \dots n \text{ terms}) \\
 &= \frac{7}{9}[(10 - 1) + (100 - 1) + (1000 - 1) + \dots n \text{ terms}] \\
 &= \frac{7}{9}[(10 + 10^2 + 10^3 + \dots n \text{ terms}) - n] \\
 &= \frac{7}{9} \left[\frac{10(10^n - 1)}{9} - n \right] \\
 &(\text{or}) \frac{70}{81}(10^n - 1) - \frac{7n}{9}
 \end{aligned}$$

34. If 7 times the 7th term of an Arithmetic Progression is equal to 11 times its 11th term, show that its 18th term is zero. Can you find the first term and the common difference? Justify your answer.

Solution:

$$7a_7 = 11a_{11}$$

$$7(a + 6d) = 11(a + 10d)$$

$$7a + 42d = 11a + 110d$$

$$a + 17d = 0$$

$$t_{18} = 0$$

Not possible because to find to unknowns a and d we need one more equations.

35. Factorize: $x^3 - 5x^2 - 2x + 24$

Solution:

$$x^3 - 5x^2 - 2x + 24$$

$$x^3 - (5x^2 - 2x + 24)$$

$$x^3 - (5x^2 - 12x + 10x - 24)$$

$$x^3 - (x(5x - 12) + 2(5x - 12))$$

$$x^3 - (5x - 12)(x + 2)$$

Now divide with $(x + 2)$

$$(x + 2)(x^2 - 3x + 4)$$

$$(x + 2)(x^2 - 4x + x - 4)$$

$$(x + 2)(x - 4)(x + 1)$$

36. Simplify: $\frac{1}{x^2 + 3x + 2} + \frac{1}{x^2 + 5x + 6} - \frac{2}{x^2 + 4x + 3}$

Solution:

$$x^2 + 3x + 2 = (x+1)(x+2)$$

$$x^2 + 5x + 6 = (x+2)(x+3)$$

$$x^2 + 4x + 3 = (x+3)(x+1)$$

$$= \frac{x+3+x+1-2(x+2)}{(x+1)(x+2)(x+3)}$$

$$= 0$$

37. If $A = \begin{pmatrix} 3 & 2 \\ -1 & 4 \end{pmatrix}$, $B = \begin{pmatrix} -2 & 5 \\ 6 & 7 \end{pmatrix}$ and $C = \begin{pmatrix} 1 & 1 \\ -5 & 3 \end{pmatrix}$ verify that $A(B + C) = AB + AC$.

Solution:

$$B + C = \begin{pmatrix} -1 & 6 \\ 1 & 10 \end{pmatrix}$$

$$A(B + C) = \begin{pmatrix} -1 & 38 \\ 5 & 34 \end{pmatrix}$$

$$AB = \begin{pmatrix} 6 & 29 \\ 26 & 23 \end{pmatrix}$$

$$AC = \begin{pmatrix} -7 & 9 \\ -21 & 11 \end{pmatrix}$$

$$AB + AC = \begin{pmatrix} -1 & 38 \\ 5 & 34 \end{pmatrix}$$

38. Find the area of the quadrilateral whose vertices are $(-4, 5)$, $(0, 7)$, $(5, -5)$ and $(-4, -2)$.

Solution: Area of the quadrilateral ABCD

$$= \frac{1}{2} \begin{vmatrix} -4 & 5 & 0 & -4 & -4 \\ -2 & -5 & 7 & 5 & -2 \end{vmatrix}$$

$$= \frac{1}{2} \{ (+20 + 35 + 0 + 8) - (-10 - 0 - 28 - 20) \}$$

$$= \frac{1}{2} (63 + 58)$$

$$= 60.5 \text{ sq. units}$$

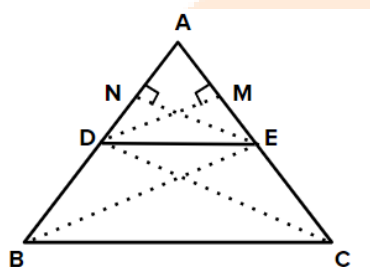
39. State and prove Thales theorem.

Solution: Thale's theorem states that if AB and C are distinct point of a triangle and two mid points joining is parallel to the third sides, so it divides the sides of triangle in corresponding motion

To prove; $= \frac{AD}{DB} = \frac{AE}{EC}$

draw $\Delta M \perp AC$ and $EN \perp AB$

ar. Of $\Delta ADE = \frac{1}{2} AD \times NE$



ar (BDE) $= \frac{1}{2} AE \times DM$ and ar (DEC) $= \frac{1}{2} EC \times DM$

$$\frac{ar(ADE)}{ar(BDE)} = \frac{\frac{1}{2} AD \times EN}{\frac{1}{2} BD \times EN} = \frac{AD}{DB} \quad \dots(1)$$

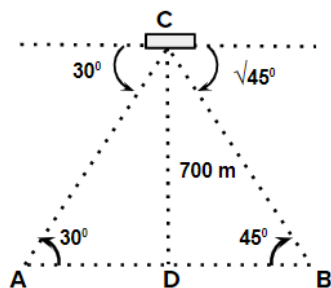
$$\frac{ar(ADE)}{ar(DEC)} = \frac{\frac{1}{2} AE \times DM}{\frac{1}{2} EC \times DM} = \frac{AE}{EC} \quad \dots(2)$$

So, from (1) and (2) $\frac{AD}{DB} = \frac{AE}{EC}$ Proved

40. A person in a helicopter flying at a height of 700 m, observes two objects lying opposite to each other on either banks of a river. The angles of depression of the objects are 30° and 45° , find the width of the river. ($\sqrt{3} = 1.732$)

Solution: width of river $AB = AD + DB$

$$\tan 30 = \frac{CD}{AD}$$



$$AD = 700\sqrt{3}$$

$$\tan 45 = \frac{CD}{DB}$$

$$DB = 700$$

$$AB = 700\sqrt{3} + 700$$

$$= 700(\sqrt{3} + 1)$$

$$= 1912.4 \text{ m}$$

41. The radii of two circular ends of a frustum shaped bucket are 15 cm and 8 cm. If its depth is 63 cm, find the capacity of the bucket in litres. (Take $\pi = \frac{22}{7}$)

Solution: $R = 15$ $r = 8$ $h = 63$ volume of frustum = volume of bucket

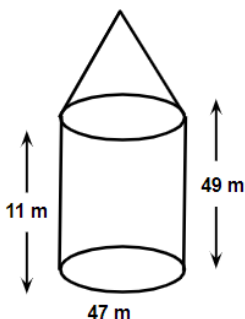
$$= \frac{1}{3}\pi h(R^2 + r^2 + R \times r)$$

$$= \frac{1}{3} \times \frac{22}{7} \times 63(15^2 + 8^2 + 15 \times 8)$$

$$= 26994 \text{ cm}^3 \text{ (or) } 26.994 \text{ lt}$$

42. A circus tent is to be erected in the form of a cone surmounted on a cylinder. The total height of the tent is 49 m. Diameter of the base is 45 m and height of the cylinder is 21 m. Find the cost of canvas needed to make the tent, if the cost of canvas is Rs.12.50 / m² (Take $\pi = \frac{22}{7}$)

Solution: Total surface area of canvas



$$= \text{CSA of cylinder} + \text{CSA of cone}$$

$$= 2\pi rh + \pi rl$$

$$= 2\pi \times 21 \times 21 + \pi \times 21 \times l$$

$$\left(l = \sqrt{21^2 + 28^2} \right)$$

$$= 5082 \text{ m}^2$$

$$\text{Cost of } 1\text{m}^2 \text{ canvas} = 12.50 \text{ Rs}$$

$$\text{Total cost} = 5082 \times 12.50$$

$$= 63525 \text{ Rs}$$

43. Calculate the standard deviation of the following data:

x	3	8	13 - a	18	23
y	7	10	15	10	8

Solution:

x	f	d	fd	d ²	fd ²
3	7	-10	-70	100	700
8	10	-5	-50	25	250
13 - a	15	0	0	0	0
18	10	5	50	25	250
23	8		80		800
	$\Sigma f = 50$		$\Sigma fd = 10$		$\Sigma fd^2 = 1000$

$$s = \sqrt{\frac{\Sigma fd^2}{\Sigma f} - \left(\frac{\Sigma fd}{\Sigma f} \right)^2}$$

$$= \sqrt{\frac{1000}{50} - \left(\frac{10}{50} \right)^2} = 6.32$$

44. A box contains 4 white balls, 6 red balls, 7 black balls and 3 blue balls. One ball is drawn at random from the bag. Find the probability that the ball drawn is:

(i) neither white nor black

(ii) red or white, and

(iii) either white or red or black or blue

Solution:

$$P(S) = 20$$

$$n(w) = 4$$

$$n(R) = 6$$

$$n(B) = 7$$

$$n(\text{Blue}) = 3$$

$$P(W(B)) = \frac{9}{20}$$

$$P(R/W) = \frac{10}{20} = \frac{1}{2}$$

$$P(\text{any}) = \frac{20}{20} = 1$$

45.

(a) If the equation $(1 + m^2)x^2 + 2mcx + c^2 - a^2 = 0$ has equal roots, then prove that $c^2 = a^2(1 + m^2)$.

OR

(b) Find the equation of the straight line segment whose end points are the points of intersection of the straight lines $2x - 3y + 4 = 0$, $x - 2y + 3 = 0$ and the midpoint of the line joining the points $(3, -2)$ and $(-5, 8)$.

Solution:

$$(a) \ a = 1 + m^2 \quad b = 2mc \quad c = c^2 - a^2$$

Since $\Delta = 0$ because roots are equal

$$= (2mc)^2 - 4(1 + m^2)(c^2 - a^2) = 0$$

$$= 4m^2c^2 - (4 + 4m^2)(c^2 - a^2) = 0$$

$$= 4m^2c^2 - \{4c^2 - 4a^2 + 4m^2c^2 - 4m^2a^2\} = 0$$

$$= m^2c^2 - c^2 + a^2 - m^2c^2 + a^2m^2 = 0$$

$$= c^2 = a^2(1+m^2)$$

$$(b) \ 2x - 3y = -4$$

$$x - 2y = -3 \times 2$$

$$= 2x - 3y = -4$$

$$2x - 4y = -6$$

on solving $x = 1, y = 2$

mid point of $(3, -2)$ and $(-5, 8)$

$$= \frac{3-5}{2}, \frac{-2+8}{2}$$

$$= (-1, 3)$$

equation of straight line

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$$

$$= y - 2 = \frac{3-2}{-1-1}(x-1)$$

$$y - 2 = \frac{1}{-2}(x-1)$$

$$-2y + 4 = x - 1$$

$$0 = x + 2y - 5$$

46.

(a) Draw the two tangents from a point which is 8 cm away from the centre of a circle of radius 3 cm. Also, measure the lengths of the tangents.

OR

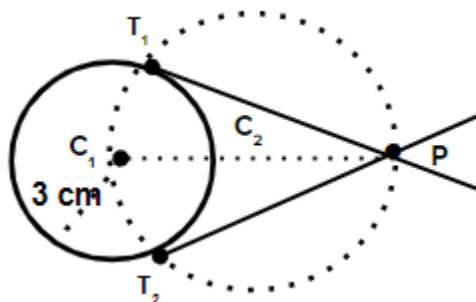
(b) Construct a cyclic quadrilateral PQRS given $PQ = 5$ cm, $QR = 4$ cm, $\angle QPR = 35^\circ$ and $\angle PRS = 70^\circ$.

(a) **Step:**

(i) draw circle c_1 of 3 cm

(ii) draw circle c_2 of 4.5 cm in b/w c_1 and P

(iii) Join T_1 and T_2 P



(b) Step:

(i) Draw $PQ = 5 \text{ cm}$

(ii) Draw $\angle QPR = 35^\circ$

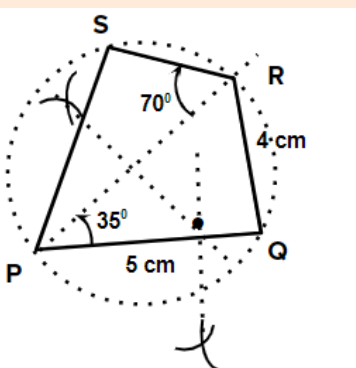
(iii) Take arc of 4 cm to get R

(iv) Join PQR

(v) Draw circumcircle by perpendicular bisector

(vi) Draw $\angle PRS = 70^\circ$

(vii) Join PQRS to get a quadrilateral



47.

(a) Draw the graph of $y = x^2 + 2x - 3$ and hence find the roots of $x^2 - x - 6 = 0$.

OR

(b)

No. of workers x	3	4	6	8	9	16
No. of days y	96	72	48	36	32	18

Draw graph for the data given in the table. Hence find the number of days taken by 12 workers to complete the work.

Solution:

For $x = 0$, $y = 0^2 - 2 \times 0 - 3 = -3$

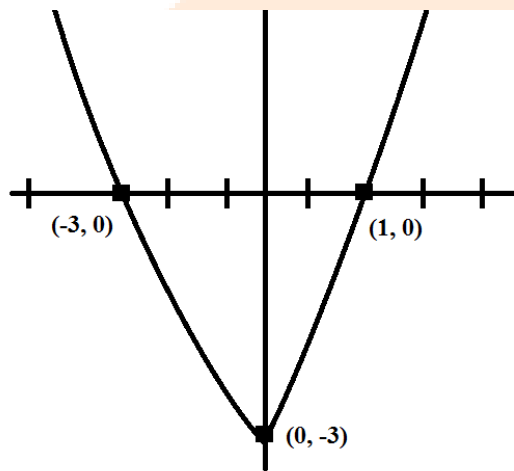
For $y = 0$, $0 = x^2 + 2x - 3$

$= x^2 + 3x - x - 3$

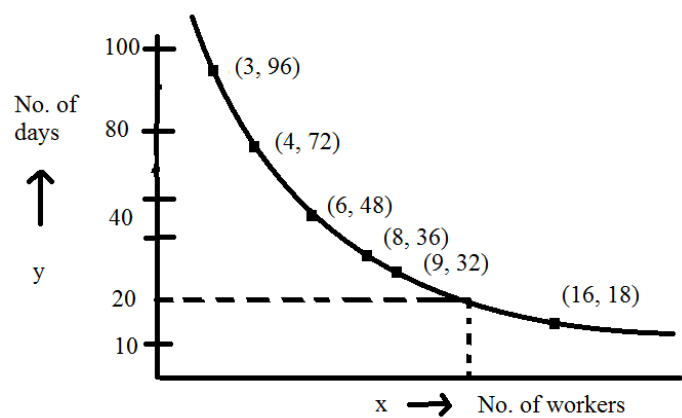
$= x(x + 3) - 1(x + 3)$

$x = -3, 1$

So, points are $(0, -3)$ $(-3, 0)$ $(1, 0)$



(b)



From figure 12 workers takes almost 20 days